Power BI Assignment 2 – DAX & Data Visualization

1. Project Overview

This project analyzes sales data using Power BI. Various visualizations have been created to identify sales trends, regional performance, customer behavior, and product distribution. DAX formulas were used to calculate important business metrics, and an interactive dashboard was designed for better decision-making.

2. DAX Calculations

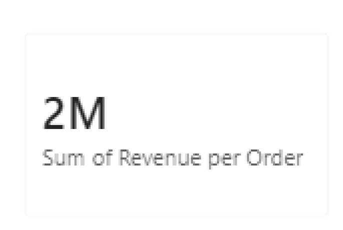
Calculated Columns:

- **Created Column for 'Category Type'**

Category Type
Clothing - Saree
Clothing - Saree
Clothing - Saree
Clothing - Saree

- **Created Column for 'Revenue Per Order' and also verified the sum of revenue by inserting in the card and got the value of 2Million**

Revenue per Order
1683
952
579
1413
525
100



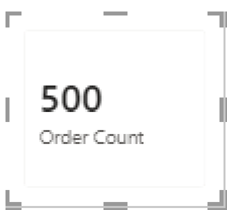
- **Created Column for 'Sales Category'**

```
Sales Category =
VAR AvgAmount =
    CALCULATE(
        AVERAGE('Order Details'[Amount]),
        ALL('Order Details')
    )
RETURN
IF(
    'Order Details'[Amount] > AvgAmount,
    "Above Average",
    "Below Average"
)
```

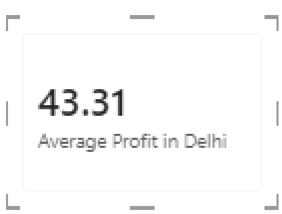
Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
561	212	3	Clothing	Saree	Clothing - Saree	1683	Above Average
119	-5	8	Clothing	Saree	Clothing - Saree	952	Below Average
193	-166	3	Clothing	Saree	Clothing - Saree	579	Below Average

3. Calculated Measures

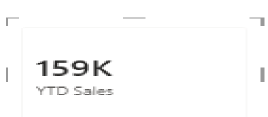
- **Created measure for Order Count and verified the received result by inserting in the Card**



- **Created measure for Average profit in Delhi and verified the received result by inserting in the Card.**

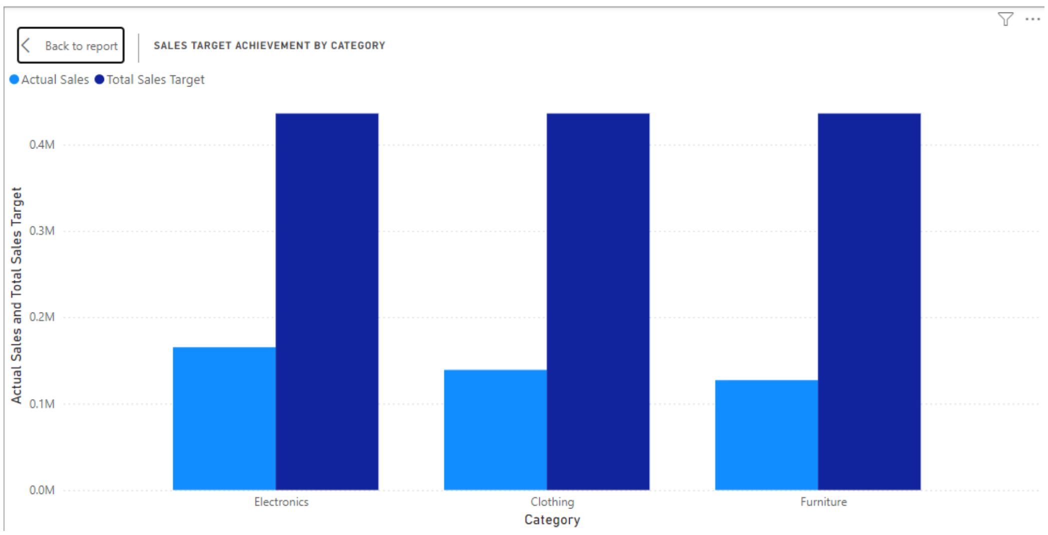


- **Created a measure to find the YTD Sales and verified the result by inserting it in the card.**



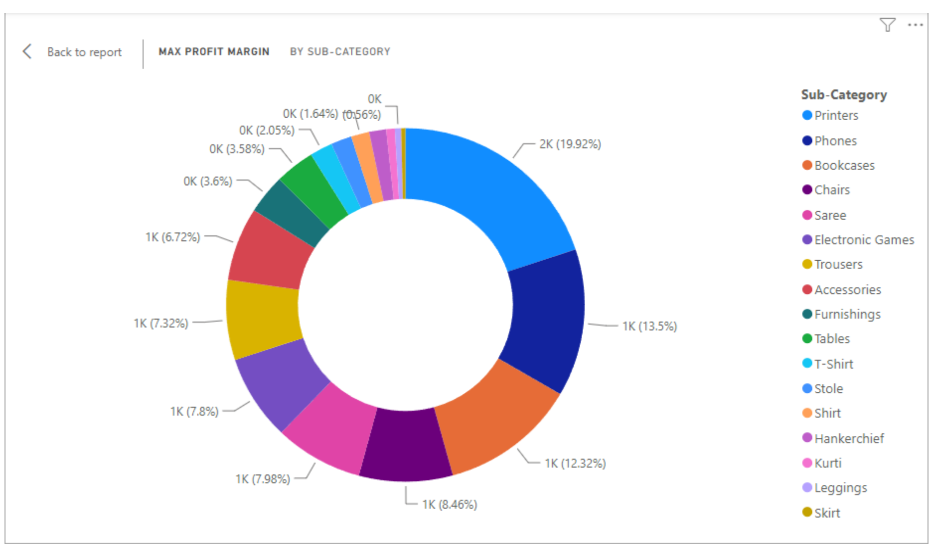
Data Visualizations

- **Sales Target Achievement by Category – Compared the actual sales with sales target and got the output. Electronics were the highest where as the Furniture category stands the lowest sales achievement.**



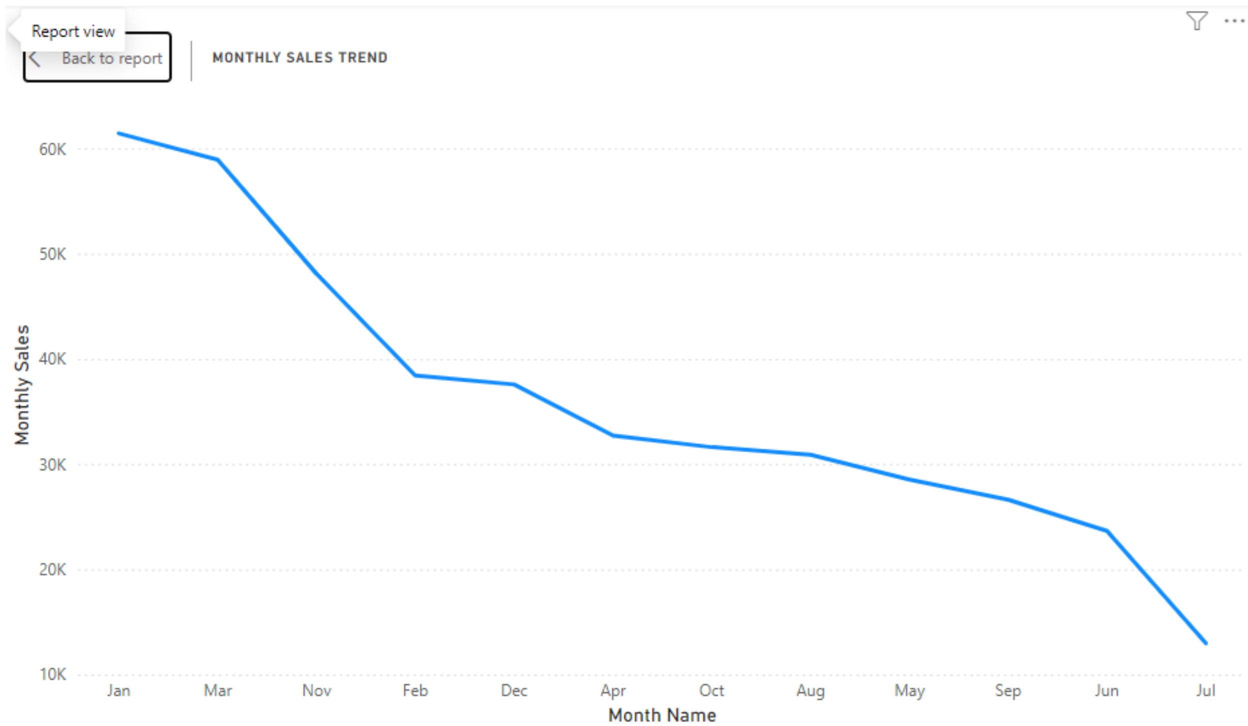
- **Max Profit Margin by Sub-Category**

  We have used the donut chart category to understand which product has most percentage of profit and got the output as Printer is the most profitable product under the Electronics category

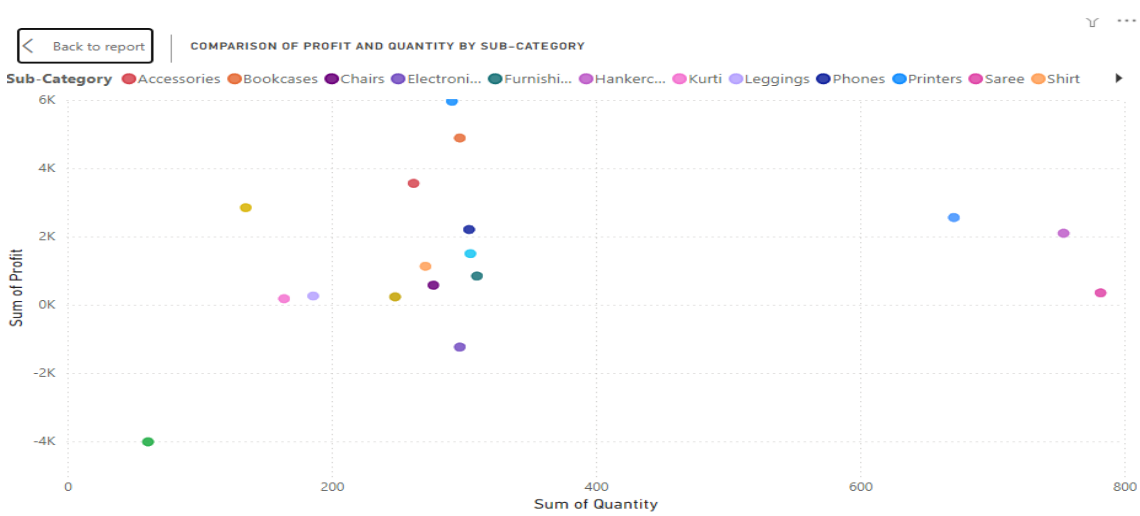


- **<u>Monthly Sales Trend</u>**

To get the output for understanding the sales trend we have used Line Chart. The sales trend shows as negative as the trend is going downwards every month.



- **<u>Comparison of Profit and Quantity by Sub-Category</u>**



- **Comparison of Total Sales Amount and Target**

Total Sales Amount

432K

Actual Sa...

Sales Target

436K

Total Sales T...

- **Comparison of Profit and Quantity by Sub-Category**

Category	Minimum Target
Clothing	12000
Electronics	9000
Furniture	10400
Total	**9000**

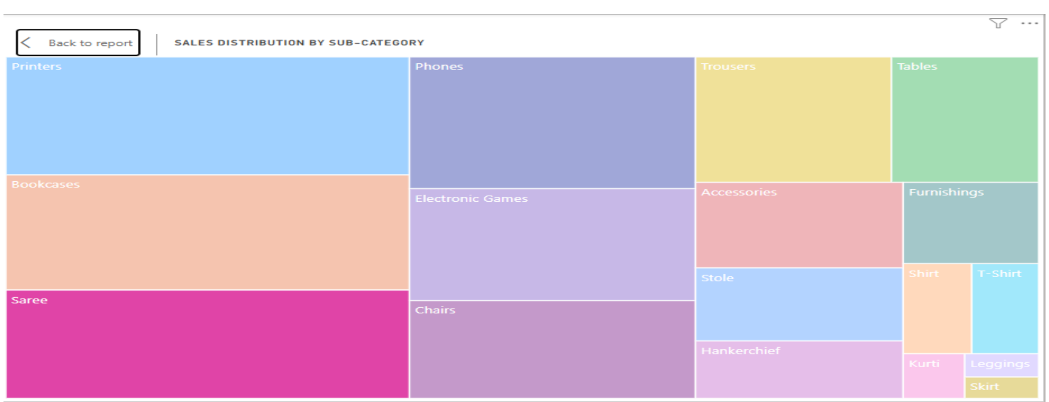
- **Sales Performance Matrix**

Back to report

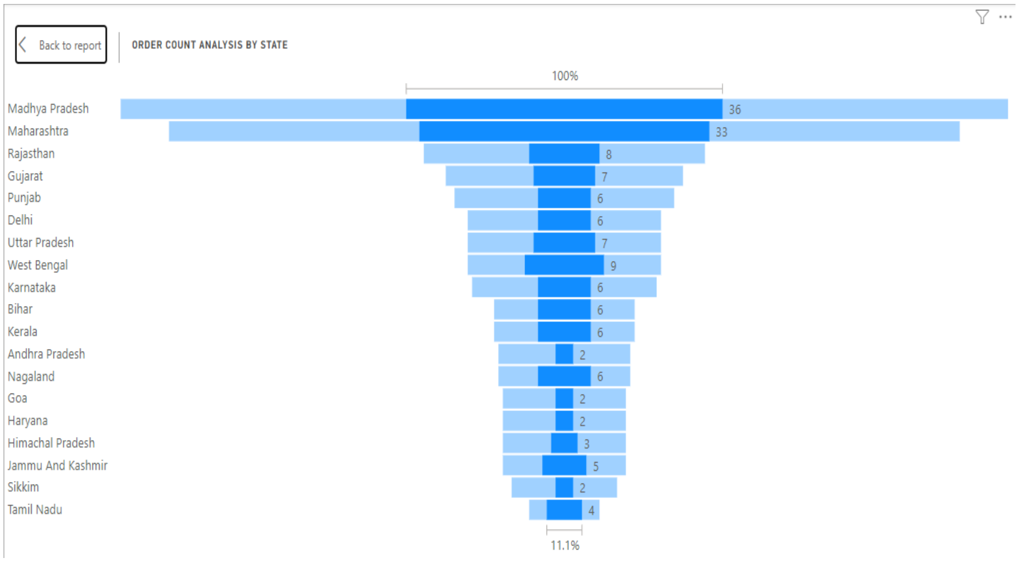
SALES PERFORMANCE MATRIX

Category	Apr	Aug	Dec	Feb	Jan	Jul	Jun	Mar	May	Nov	Oct	Sep	**Total**
Clothing	31400	33900	36400	43600	43500	33800	31600	43800	31500	36300	36100	34000	**435900**
Electronics	31400	33900	36400	43600	43500	33800	31600	43800	31500	36300	36100	34000	**435900**
Furniture	31400	33900	36400	43600	43500	33800	31600	43800	31500	36300	36100	34000	**435900**
Total	**31400**	**33900**	**36400**	**43600**	**43500**	**33800**	**31600**	**43800**	**31500**	**36300**	**36100**	**34000**	**435900**

- **Sales Distribution by Sub-Category**



- **<u>Order Count Analysis by State</u>**



<u>Key Insights</u>

- Electronics achieved the highest sales performance, while Furniture recorded the lowest sales achievement.
- Printers generated the highest profit margin among all product sub-categories.
- The monthly sales trend shows a decline in sales over the observed period.
- Sales and profit vary across sub-categories, indicating that higher sales quantity does not always result in higher profit.
- The treemap highlights the major contributors to overall sales across different sub-categories.
- The state-wise order analysis identifies regions with higher customer order volumes.
- The dashboard enables quick monitoring of sales, targets, profitability, and regional performance, supporting better business decisions.

<u>Conclusion</u>

The dashboard provides valuable insights into sales performance, profitability, product trends, and regional demand. By combining DAX calculations with interactive Power BI visualizations, it helps stakeholders monitor business performance, identify high-performing categories, track sales targets, and make data-driven decisions.